

 FIR/Testing-Outdoor/518300/2023/6085	FIR VEHICLE TESTING	Date: 08/11/2023		
Part No / Group: 51832155R	Specification no :- " 259051830006 "		Action by	Failure class

Subject: -	Report on fluid economy evaluation of TML Signa 2820.T CNG Cx vehicle fitted with TATA 5.7 SGI BS-VI OBD-II (200 HP) engine, G750 (FGR:9.19) Transmission and 6.43 Rear axle ratio in comparison with TML Signa 2818.T Ashwamedh BS-VI OBD-I vehicle as a Base vehicle on JSR-Ranchi highway				
1.0 Conclusion:					
1.1 Fluid economy in laden(rated load) condition at 40 kmph :					
Fluid economy of Signa 2820.T CNG Cx BS-VI OBD-II vehicle was found 4.57 kmpl which was superior by 13.17 % wrt Signa 2818.T BS-VI OBD-I Base vehicle					
1.2 Fluid economy in laden(rated load) condition at 55 kmph :					
Fluid economy of Signa 2820.T CNG Cx BS-VI OBD-II vehicle found 5.08 kmpl which was superior by 32.28 % wrt Signa 2818.T BS-VI OBD-I Base vehicle					
1.3 Fluid economy in Unladen condition at 40 kmph :					
Fluid economy of Signa 2820.T CNG Cx BS-VI OBD-II vehicle found 6.75 kmpl which was inferior by 3.63 % wrt Signa 2818.T BS-VI OBD-I Base vehicle					
1.4 Fluid economy in Unladen condition at 55 kmph :					
Fluid economy of Signa 2820.T CNG Cx BS-VI OBD-II vehicle found 6.35 kmpl which was which was superior by 8.13% wrt Signa 2818.T BS-VI OBD-I Base vehicle					
1.5 Overall fluid economy status at 40 kmph (80% Laden + 20% Unladen condition):					
- Fluid economy of Signa 2820.T CNG Cx BS-VI OBD-II vehicle found 4.89 kmpl which was inferior by 12 % wrt Signa 2818.T BS-VI OBD-I Base vehicle - Signa 2820.T CNG Cx BS-VI OBD-II meets the PALS target of “Leadership”.					
1.6 Overall fluid economy status at 55 kmph (80% Laden + 20% Unladen condition):					
Fluid economy of Signa 2820.T CNG Cx BS-VI OBD-II vehicle found 5.29 kmpl which was Superior by 28 % wrt Signa 2818.T BS-VI OBD-I Base vehicle					
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2.0 Result: Vide Observations		
FAILURES CLASS: A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement. B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention. C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule. D: Customer perception issues and minor irritants.		
3.0 Test Vehicles: - 1. Signa 2820.T CNG BS-VI OBD-II (EJ00789)_Test vehicle Chassis no. - PC82553032RL10006; Engine no. - HXX814549. 2. Signa 2818.T BS-VI OBD-I (ERCJ 1124)_Base vehicle Chassis no. - PC13609091RJ00632; Engine no. - 91H63804924. Input Doc Ref.: - Test Request No :- Testing-Outdoor/15523 DID No :- NA. DDP No: - NA.	Enclosures: Annexure 1: Panel chart Annexure 2: PAT sheet Annexure 3: Duty cycle report Annexure 4: Test request Annexure 5: Test plan	
	Work Order No. :- P.22.EA.501	Make :- TATA Motors Limited
Earlier Ref: - Nil.		
4.0 Project Status:- DR3		
5.0 Introduction: As a part of RDE (OBD-II) Testing Program, TML Signa 2820.T CNG Cx vehicle fitted with TATA 5.7 SGI BS-VI OBD-II (220 HP) engine, G750 (FGR:9.19) Transmission and 6.43 Rear axle ratio was offered for fluid economy evaluation along with TML Signa 2818.T BS-VI OBD-I vehicle as a base vehicle. All vehicles were tested back-to-back at JSR- Ranchi-JSR route in laden and unladen conditions at 40 km/hr as well as 55 km/hr vehicle cruising speeds. The subject report will provide detail about fluid economy evaluation test carried out on all vehicles (Test vehicle, Base vehicle).		
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Part No / Group: 51832155R	Specification no :- " 259051830006 "		Action by	Failure class

6.0 Testing:

Following test was conducted during evaluation :

Sr. no.	Test	Reference standard	Instruments used
1	Fluid economy evaluation	TST/VT/WG/158	Cummins Data Logger

7.0 Vehicle details:

Parameter		Base Vehicle	Test vehicle
Vehicle model		Signa 2818.T (OBD-I) (ERC-1183)	Signa 2820.T CNG (OBD-II) (ERC-1219)
Engine Model		Cummins Ashwamedh 5.6L BS-VI (180 HP) OBD-I	TATA 5.7 SGI BS-VI (200 HP) OBD-II
Max Power		180 HP @2300 RPM	200 HP @2250 RPM
Max Torque		850 Nm @ 1000 ~1600 rpm	700 Nm @ 1300 ~1600 rpm
GVW (kg)		28000	28000
Gear Box		Model: Tata G950, Type: DD	Model: Tata G750, Type: DD
Gear Ratios.		1 st gear:9.36, 2 nd gear: 5.2, 3 rd gear: 3.05, 4 th gear: 1.84, 5 th gear: 1.33, 6 th gear: 1	1st- 9.19, 2nd- 4.98, 3rd- 2.96, 4th- 1.93, 5th- 1.40, 6th- 1.00
Axle Ratio		4.88	6.43
Tyres	FA1	Size: 295/90R20, Make: M/s Bridgestone , Type: Regular (Non-LCRR)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
	FA2	Size: 295/90R20, Make: M/s Bridgestone , Type: Regular (Non-LCRR)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
	LA1	Size: 295/90R20, Make: M/s Bridgestone , Type: Regular (Non-LCRR)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
	RA1	Size: 295/90R20, Make: M/s Bridgestone , Type: Regular (Non-LCRR)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
	RA2	Size: 295/90R20, Make: M/s Bridgestone , Type: Regular (Non-LCRR)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
Tractive effort @ 1 st gear		74804 N	79700 N
Tractive effort @ Top gear		7992 N	8672 N
Fan Details (Type/Make/Blades)		Type: E-Viscous fan and Size:24"	Type: Viscous fan and Size:22"

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Part No / Group: 51832155R	Specification no :- " 259051830006 "		Action by	Failure class

Parameter	Base Vehicle	Test vehicle
Vehicle model	Signa 2818.T (OBD-I) (ERC-1183)	Signa 2820.T CNG (OBD-II) (ERC-1219)
Base Cal Improvement (Cal Maturity %)	As per production released	95% Cal Maturity
Low Viscous Engine Oil (TMGO- 10W30)	No (15W40CK4+)	No (15W40CK4+)
FE Switch	Yes (3 Mode : E/B/P)	Yes (2 Mode : E/P)
Engine Features (EOL : Enable/Disable)	Enable	Enable
ATS Optimization	No	No
APM	No	No
Cruise Control	No	No
Gear based VAM	No	No
Intelligent mode shifter	No	No
Moving Mass Estimation	No	No
Driveline Optimization	NA	Yes
GSA (Gear Shift Advisor)	Yes	No
MSA (Mode Shift Advisor)	NA	No
Low CRR Tyres	No	Yes
Low Viscous rear axle oil (80W90)	No	No
Smart Alternator (LIN)	NA	No
THU/Unitised/Low friction bearing	Yes	Yes

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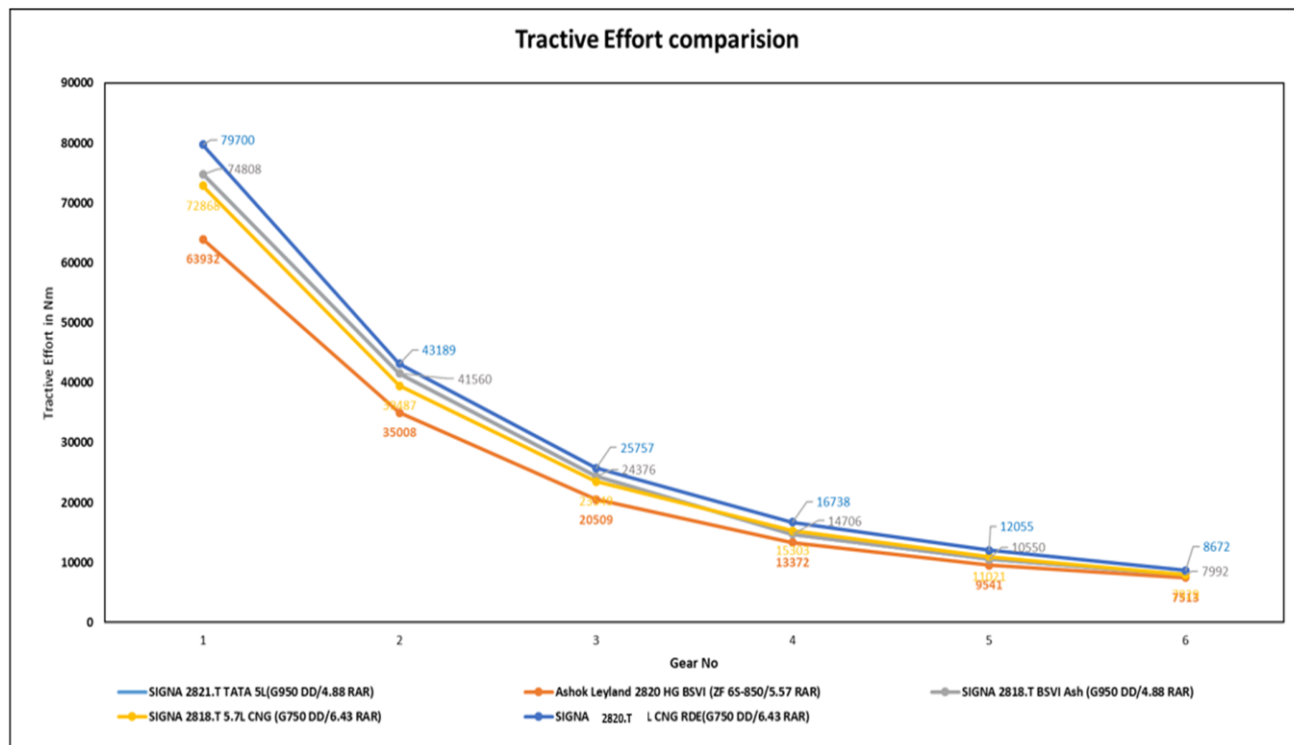
8.0 Test result:

8.1 Cruising speed: 40 km/hr & 55 Km/hr

Sl. No.	Parameters			Base Vehicle				Test Vehicle												
				SIGNA 2818.T Ashwamedh BSVI				Signa 2820.T CNG BSVI RDE												
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)			ERC-1124				EJ00789												
2	Vehicle GVW/GCW		kg	28000				28000												
	Unladen weight		kg	--				7550												
3	Engine Details (Make/cc/Norms/no. of Cyl.)			Ashwamedh 5.6L				TATA 5.7 SGI Bharat Stage VI OBD-II												
	Engine Max Power			180 HP @ 2300 rpm				200 HP @ 2250 rpm												
	Engine Max Torque			850 Nm @ 1000-1600 rpm				700 Nm @ 1300 - 1600 rpm												
	Engine Calibration No.			KT10008.17				57SGITCIBS6												
4	Engine Calibration Maturity Level			As per production released				Develop Cal												
	Regeneration Strategy (Time ase/Soot base)			Time Based (100 Hrs)				NA												
5	Gear Box (make/drive/FGR)			G950 6S DD, FGR- 9.36				G750 6S DD, FGR- 9.19												
	Gear Ratios			1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00				1st- 9.19, 2nd- 4.98, 3rd- 2.96, 4th- 1.93, 5th- 1.40, 6th- 1.00												
6	RAR			4.88				6.43												
7	Tyre Details			Bridgestone 295/90R20 Non low CRR				Bridgestone 295/90R20 Low CRR												
8	LBSC	High & Low Engine Speed Breakpoint	RPM	1900 rpm,1900 rpm				NA												
		Vehicle Mass Threshold	kg	28000				NA												
	VAM	Variable Acceleration 1, 2	kmph/sec	2.5,2.5				NA												
		Variable Acceleration Speed 1, 2	kmph	10, 40				NA												
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph	50, 50				NA												
	RSG	Maximum Accelerator Vehicle Speed	kmph	80				74												
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance	100%				--												
		Torque Limit Maximum Adjustment Factor	Economy	100%				--												
	PTP	Powertrain Protection (Enable / Disable)		Disable				NA												
		Torque Limit		NA				NA												
9	USFE / Torque Table Revision (1/2/3/n)			USFE-1				USFE-1												
	Fan Details (type/make/blades)			24" E-Viscous fan				22" Viscous fan												
	Base Cal Improvement (Cal Maturity %)			As per production released				100% Cal Maturity												
	Low Viscous Engine Oil (TMGO-10W30)			No (15W40CK4+)				No (15W40CK4+)												
	FE Switch			Yes (3 Mode : E/B/P)				Yes(Eco & Power)												
	Engine Features (EOL : Enable/Disable)			Enable				No												
	ATS Optimization			No				No												
	APM			No				No												
	Gear based VAM			No				No												
	Intelligent mode shifter			No				No												
	Moving Mass Estimation			No				No												
	Driveline Optimization			NA				NA												
	GSA (Gear Shift Advisor)			Yes				Yes (Not Tuned)												
	MSA (Mode Shift Advisor)			NA				NA												
	FE Avenues (Vehicle Level)	Low CRR Tyres			No				Yes											
Variable displacement pump			NA				NA													
Low Viscous rear axle oil (80W90)			No				No													
Smart Alternator (IUN)			NA				NA													
THU/Unitised/Low friction bearing			No				Yes (Except RA1)													
Route			Jamshedpur-Ranchi-Jamshedpur				Jamshedpur-Ranchi-Jamshedpur													
FE Trial Details		FE Duty Cycle (Laden/Unladen)			Laden Trial @ 40 kmph		Unladen Trial @ 40 kmph		Laden Trial @ 55 Kmph		Unladen Trial @ 55 Kmph		Laden Trial @ 40 kmph		Unladen Trial @ 40 kmph		Laden Trial @ 55 Kmph		Unladen Trial @ 55 Kmph	
		Trial Date			13.10.23	15.10.23	01.10.23	10.10.2023	11.10.2023	21.10.23			13.10.23	15.10.23	01.10.23	10.10.2023	11.10.2023	21.10.23		
		Weather condition			Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny
		Traffic Condition (Light/Average/Heavy)			Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average
	Trip Distance			241.00	241.00	241.00	241.00	241.00	241.00	241.00	241.00	241.00	182.80	183.00	270.00	204.00	202.60	254.00	254.00	
	Total CNG quantity filled			Kg		40.00		40.00		40.00		40.00		40.00		40.00		40.00		
	Total Diesel Quantity			Litre		58.91	57.50	35.70	61.20	61.00	40.04									
	Total Urea Quantity			Litre		2.8	3.0	2.6	3.0	3.5	2.0									
	Equivalent Urea Quantity (50% Consumption)			Litre		1.40	1.50	1.30	1.50	1.75	1.00									
	Total Equivalent Fluid Consumed			Litre		60.31	59.00	37.00	62.70	62.75	41.04									
	Urea uses % wrt. Diesel			%		4.75%	5.22%	7.28%	4.90%	5.74%	5.00%									
	Fuel Economy			Kmpl		4.09	4.19	6.75	3.94	3.95	6.02									
	Equivalent Fluid Economy			Kmpl/Kmpkg		4.00	4.08	6.51	3.84	3.84	5.87	4.57	4.58	6.75	5.10	5.07	6.35			
	Status of Equivalent Fluid economy			%		Base	Base	Base	Base	Base	Base	14.36%	12.00%	3.63%	32.68%	31.88%	8.13%			
	Cluster Total Odo. Reading (at End of trial)			Km		251	251	250	251	251	250	194.5	196.2	272.2	207.0	205.6	256.6			
Cluster Trip AFE			Kmpl/Kmpkg		4.50	4.50	6.90	4.30	4.10	6.70	4.6	4.5	7.6	4.6	4.4	6.5				
Status of Cluster Trip AFE wrt Actual fluid economy			%		9.09%	6.86%	2.16%	8.42%	3.64%	10.16%	0.65%	-1.67%	11.18%	-10.87%	-15.11%	2.31%				
Avg. Equivalent Fluid Economy				4.04		6.51		3.84		5.87		4.57		6.75		5.08		6.35		
Status of Avg. Fluid Economy						Base						13.17%		3.63%		32.28%		8.13%		
Overall	Overall Avg. Fluid Economy (kmpl) : L + UL			80%	20%	4.37			4.13			4.89			5.29					
	Status of Overall Avg. Eq. Fluid Economy (%)			Base			Base			12%			28%							

 FIR/Testing-Outdoor/518300/2023/6085		FIR VEHICLE TESTING		Date: 08/11/2023					
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Action by	Failure class								

8.3 Tractive Effort Comparison :



Remarks:

- Signa 2820.T CNG Cx BS-VI OBD-II (220 HP) Test vehicle tractive effort was better in all gears as compared with Signa 2818.T BS-VI OBD-I (180 HP) Base vehicle.

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Action by	Failure class						

8.4 Torque Table :

8.3.1: Test vehicle (Signa 2820.T CNG BS-VI OBD-II (200 HP))

➤ Power mode – Test Vehicle

float y \ x	800	1000	1200	1400	1600	1800	2000	2200
-1	445	545	680	700	700	690	670	630
0	445	545	680	700	700	690	670	630
1	445	545	680	700	700	690	670	630
2	445	545	680	700	700	690	670	630
3	445	545	680	700	700	690	670	630
4	445	545	680	700	700	690	670	630
5	445	545	680	700	700	690	670	630
6	445	545	680	700	700	690	670	630

➤ Light mode – Test Vehicle

float y \ x	800	1000	1200	1400	1600	1800	2000	2200
-1	445	545	680	700	700	690	670	630
0	445	545	680	700	700	690	670	630
1	431	530	670	701	701	683	663	630
2	431	530	670	701	701	683	663	630
3	445	530	540	540	491	442	442	442
4	445	530	503	503	458	412	412	412
5	445	410	410	410	410	365	365	365
6	445	432	432	432	432	384	384	384

9.0 Observation: Nil

(Failure and Mint ID)

10.0 Further Testing: Further fluid economy evaluation needs to be done with benchmark vehicle to get the PALS attribute wrt core competitor.

FIR Status

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Annexure-1

Level 1 Attribute:

PAT- Fuel Economy

Programme / MY : SIGNA 2820.T CNG Cx BSVI RDE

Gateway : DR3

PANEL Chart

Specification:Max Power-200 hp @ 2250 rpm
Max Torque - 700 Nm 1300-1600 rpm
Driveline : G-750 DD | 6.43 RAR | 295/90R 20 Tyres | 2 Mode FE Switch |
EOL Parameters : RSG: 67 kmph | PTP: Disable

Owner: Ajaykumar Jadhav

Project Name: SIGNA 2820.T CNG BSVI RDE(Cx Version)

VE CVBU/FE/01

Panel Chart No: VE CVBU/ HCV/ SIGNA 2820.T CNG Cx BSVI RDE/ Cx / FE/01

Date: 2/11/2023

Level 1 Summary:

Level 1:

PALS Position

L

Status

Current/
Gateway

L

Job#1 Predicted

Statement Of Intent (SOI) or DNA Description:

Fuel Economy: Haulage Application

Attribute Target Summary:

Deriv.

Level 2

Level 3

Level 4

Level 5

Level 6

PALS

Metric

Target

Status

Current/Gateway
SIGNA 2820.T CNG Cx BSVI RDE

Job#1
Predicted

Base Vehicle

Base
Signa 2818.T BSVI

LBSC - 1900.1900.28000 Kg
VAM- 2.5 Kmph/s, 2.5Kmph/s, 10 kmph,
40kmph
GDP- 50 kmph, 50 kmph
RSG - 80kmph

Remarks

Specification

Vehicle GVW (Payload)

kg

28000

28000

Power (Max) @ rpm

HP

200 hp @ 2250 rpm

180 hp @ 2300 rpm

Power to Weight ratio

HP/Ton

7.14

6.42

Torque (Max) @ rpm

Nm

700 Nm @ 1300-1600 rpm

850 Nm @ 1000-1600 rpm

Torque to Weight ratio

Nm/Ton

25.00

30.35

Tyres size

--

Bridgestone 295/90R20

Bridgestone 295/90R20

Gear Box Ratios

--

G750 DD, FGR -9.19

G950 DD, FGR -9.36

Final Drive Ratio

--

6.43

4.88

Fluid Economy at Rated Load

L

L

Highway AC OFF

--

Route : Jamshedpur-Ranchi

--

Avg. Fluid Economy @40 kmph

kmpl

≥ 4.04

4.57

4.04

*Signa 2820.T CNG OBII Cx and Signa 2818.T OBDI was tested back to back condition on JSR-Ranchi route.

Avg. Fluid Economy @55 kmph

kmpl

-

5.08

3.84

Fluid Economy At Unladen

C

C

Highway AC OFF

--

Route : Jamshedpur-Ranchi

--

Avg. Fluid Economy @40 kmph

kmpl

≥ 6.51

6.75

6.51

*Signa 2820.T CNG OBII Cx and Signa 2818.T OBDI was tested back to back condition on JSR-Ranchi route.

Avg. Fluid Economy @55 kmph

kmpl

-

6.35

5.87

Overall Fuel Economy (80% Laden + 20% Unladen)

L

L

Highway AC OFF

--

Route : Jamshedpur-Ranchi

--

Overall Avg. Fluid Economy @40 kmph

kmpl

≥ 4.37

4.89

4.37

*Signa 2820.T CNG OBII Cx and Signa 2818.T OBDI was tested back to back condition on JSR-Ranchi route.

Overall Avg. Fluid Economy @55 kmph

kmpl

-

5.29

4.13

Executive Summary:

1. SIGNA 2820.T CNG Cx BSVI OBD-2 & SIGNA 2818.T BSVI OBD-1 tested back to back on Jamshedpur-Ranchi route at 40 & 55 kmph cruising speed in laden & unladen condition.

2. Test vehicle SIGNA 2820.T CNG Cx BSVI OBD-2 tested with : 200 HP Engine calibration with 95% Cal Maturity + Low viscous engine oil + Low viscous 80W90 rear axle oil +22 inch Viscous fan

3. Base vehicle SIGNA 2818.T BSVI tested with production released configuration.

4. Target has been taken at-par (As per Cx approach) wrt to base SIGNA 2818.T OBD-1 BSVI vehicle.

5. SIGNA 2820.T Cx RDE BSVI is meeting the PALS target of "Leadership".

Tested by:

Debabrata Das/

Amit Kumar Satapathy

/ Swapnil Jadhav

PAT ENGINEER

Verified and Approved by:

Ajaykumar Jadhav

/ Sanjeev Kumar

PAT LEAD / PROJECT LEAD

Enclosed with FIR/Testing-Outdoor/518300/2023/6085

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VE-CV/PanelChart/01/July18

Annexure -2

 TATA MOTORS LTD. ERC,PUNE		PAT ATTRIBUTES SHEET - Fuel Economy Project Name: SIGNA 2820.T CNG BSVI RDE_Cx (G-750 DD FGR-9.19, 6.43 RAR, 295/90R 20 phase II low CRR Tyres, 2 Mode FE Switch) Base Vehicle: SIGNA 2818.T 5.6L BSVI OBD-1_G-950 DD, FGR-9.36, 4.88 RAR, 295/90R 20 Tyres Non low CRR, 3 Mode FE Switch Competitor Vehicle: Application: Haulage										VE CVBU/FE/01 PAT Sheet No.: VE CVBU/ HCV/ SIGNA 2820.T CNG BSVI RDE/ Cx / FE/01																																			
												Date: 2/11/2023																																			
Depart ment Name: Vehicle Engineering																																															
Sr.No.	Aggregate and test parameter						Unit	Test Type	Classification @DR4	Ref.Standard / Work Guideline	Simulation	Base Vehicle SIGNA 2818.T BSVI OBD-1 (G 950 DD RAR 4.88 295/90R20)	Target	Test Vehicle SIGNA 2820.T CNG BSVI OBD-II Cx (G 750 DD RAR 6.43 295/90R20)	Status (RYG)	Remarks	CoC Remark																														
	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7																																									
A	Fuel Economy Evaluation						Heading																																								
2.0	Fuel Economy At Rated Load						Heading																																								
2.7	Highway AC OFF						Heading																																								
2.7.1	Route : Jamshedpur-Ranchi																																														
2.7.11	Average Fluid Economy @40 kmph						Kmpl	Standard			TST/VTWG/158		4.04	≥ 4.04	4.57	G	*Signa 2820.T CNG OBD Cx and Signa 2818.T OBDI was tested back to back condition on JSR-Ranchi route.																														
2.7.12	Average Fluid Economy @55 kmph						Kmpl					3.84	-	5.08	-																																
4.0	Fuel Economy At Unladen						Heading																																								
4.7	Highway AC OFF						Heading																																								
4.7.1	Route : Jamshedpur-Ranchi																																														
4.7.11	Average Fluid Economy @40 kmph						Kmpl	Standard			TST/VTWG/158		6.51	≥ 6.51	6.75	G	*Signa 2820.T CNG OBD Cx and Signa 2818.T OBDI was tested back to back condition on JSR-Ranchi route.																														
4.7.12	Average Fluid Economy @55 kmph						Kmpl					5.87	-	6.35	-																																
5.0	Overall Fuel Economy (80% Laden + 20% Unladen)						Heading																																								
5.1	Highway AC OFF						Heading																																								
5.1.1	Route : Jamshedpur-Ranchi																																														
5.1.11	Overall Average Fluid Economy @40 kmph						Kmpl				TST/VTWG/158		4.37	≥ 4.37	4.89	G	*Signa 2820.T CNG OBD Cx and Signa 2818.T OBDI was tested back to back condition on JSR-Ranchi route.																														
5.1.12	Overall Average Fluid Economy @55 kmph						Kmpl					4.13	-	5.29	-																																
<table border="1"> <tr> <th colspan="3">Sign Off</th> <th colspan="3">Index for Status</th> </tr> <tr> <th>Responsibility</th> <th>Name</th> <th>Signature</th> <th colspan="5">Off target, no resolution</th> </tr> <tr> <td>PAT Engineer</td> <td>Debabrata Das/Amit Satapathy/ Swapnil Jadhav</td> <td>  </td> <td colspan="5">Marginally off target, resolution planned/low risk</td> </tr> <tr> <td>PAT Lead / Project Lead</td> <td>Ajaykumar Jadhav / Sanjeev kumar</td> <td> </td> <td colspan="5">Met target, no concern</td> </tr> </table>																		Sign Off			Index for Status			Responsibility	Name	Signature	Off target, no resolution					PAT Engineer	Debabrata Das/Amit Satapathy/ Swapnil Jadhav	  	Marginally off target, resolution planned/low risk					PAT Lead / Project Lead	Ajaykumar Jadhav / Sanjeev kumar	 	Met target, no concern				
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PAT Lead / Project Lead	Ajaykumar Jadhav / Sanjeev kumar	 	Met target, no concern																																												

OUR CULTURE CREDO

AT TATA MOTORS

We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.

We are **MORE WHEN ONE!**

Fuel Economy Evaluation on Signa 2820.T 5.6L CNG RDE BSVI with Signa 2818.T 5.6L Cummins BSVI Base Vehicle.

PED - HCV | 07.11.2023

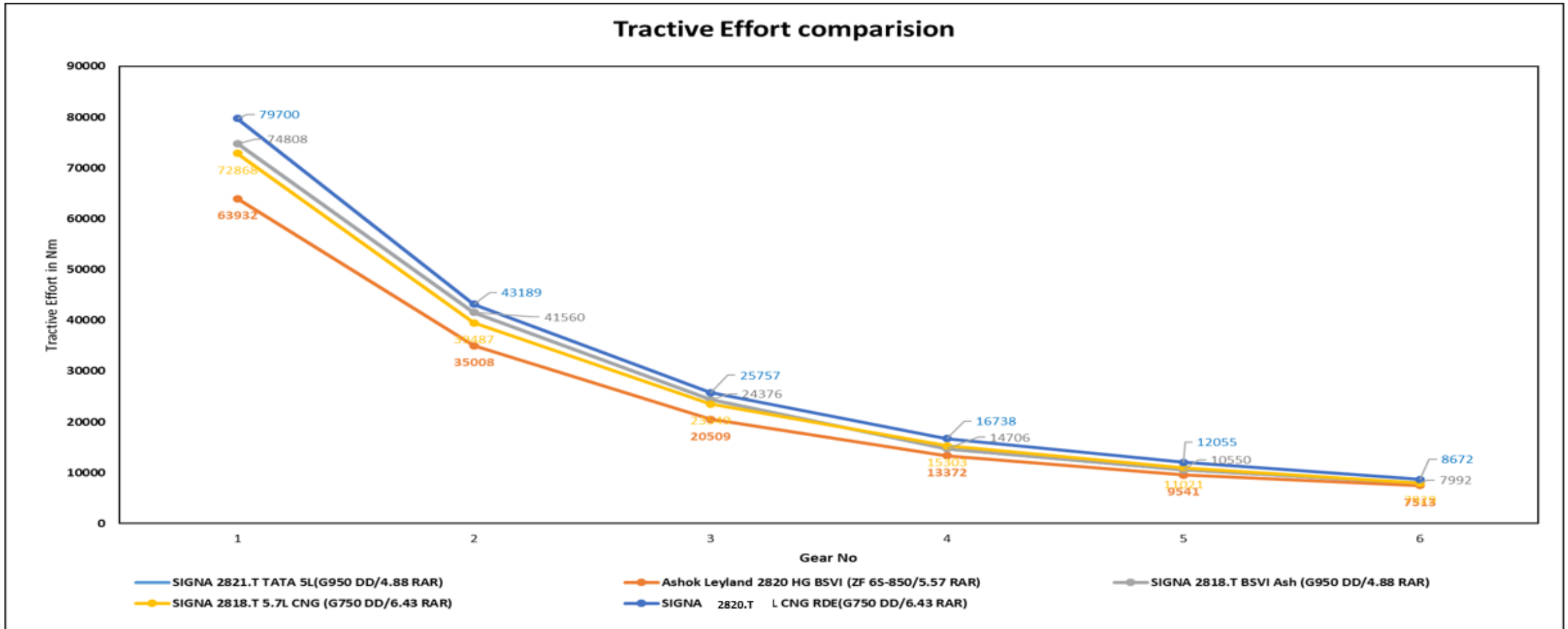


Vehicle Specification

Annexure -3

Sr. No.	Parameters	Signa 2818.T	Signa 2820.T CNG
1	Engine	Ashwamedh 5.6L	TATA 697 5.7L SGI CNG OBD II
1.1	No of cylinders	6	6
1.2	Engine Max. Power	180 HP @ 2300 rpm	204 HP @ 2250 rpm
1.3	Engine Max. Torque	850 Nm @ 1000 - 1600 rpm	700 Nm @ 1500 rpm
1.4	Power to weight (HP/Ton)	6.43	7.28
1.5	Torque to weight (Nm/Ton)	30.36	25.00
1.6	Engine Calibration Number	Production Released	--
1.7	Engine Calibration Maturity	Production Released	Develop Cal
1.8	Emission Calibration Maturity	100%	Develop
1.9	Regeneration Strategy	SCR	--
1.9.1	Exhaust Pipe Insulation	Fully wrapped metal insulation	--
2	Fuel Economy Improvement Technologies		
2.1	Engine Cooling Fan	E-Viscous (24" Fan)	E-Viscous (24" Fan)
2.2	Fuel Economy Switch	3 mode (EBP)	2 mode (EP)
3	Fuel Economy Improvement Technologies		
3.1	Fuel Tank type	Polymer	Metal Cylinder
3.2	Driveline & Ratios	G950 6S DD, FGR- 9.36	G750 6S DD, FGR- 9.19
3.3	Gear Ratio	1st: 9.36, 2nd: 5.20, 3rd: 3.04, 4th: 1.83, 5th: 1.32, 6th: 1, Rev:8.36	1st: 9.19, 2nd: 4.98, 3rd: 2.97, 4th: 1.93, 5th: 1.39, 6th: 1, Rev:8.36
3.4	RAR	4.88	6.43
3.5	Tire size	295/90R20	295/90R20
4	EOL Parameters		
4.1	LBSC, VAM, GDP	LBSC - 1900rpm, 1900rpm, 28000kg, VAM- 2.5kmph/Sec, 2.5kmph/Sec, 10kmph, 40kmph, GDP- 50kmph, 50kmph,	NA
4.2	RSG	80	74

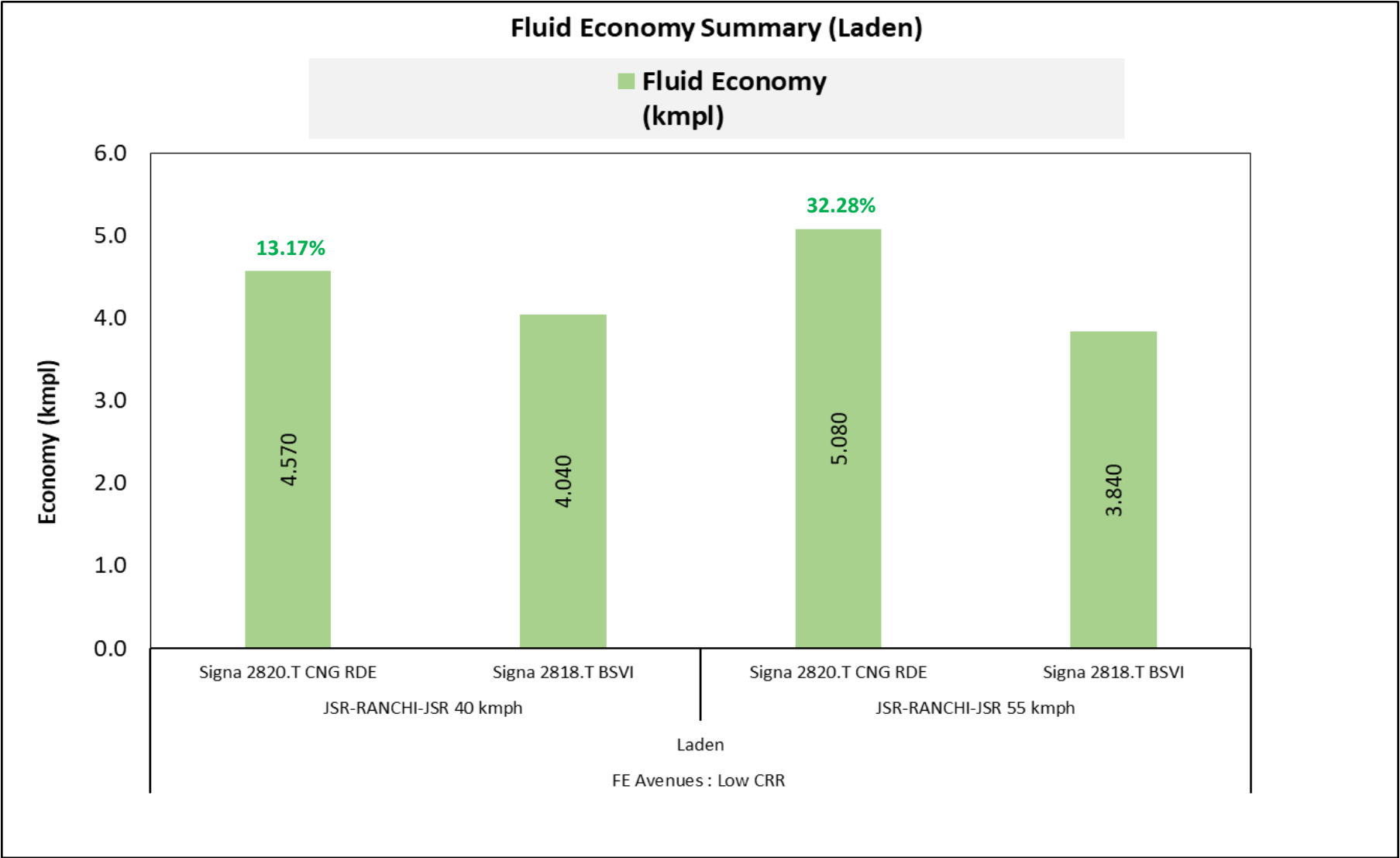
Tractive Effort Comparison



➤ Tractive effort of Signa 2820.T CNG RDE is better as compared to Signa 2818.T BSVI

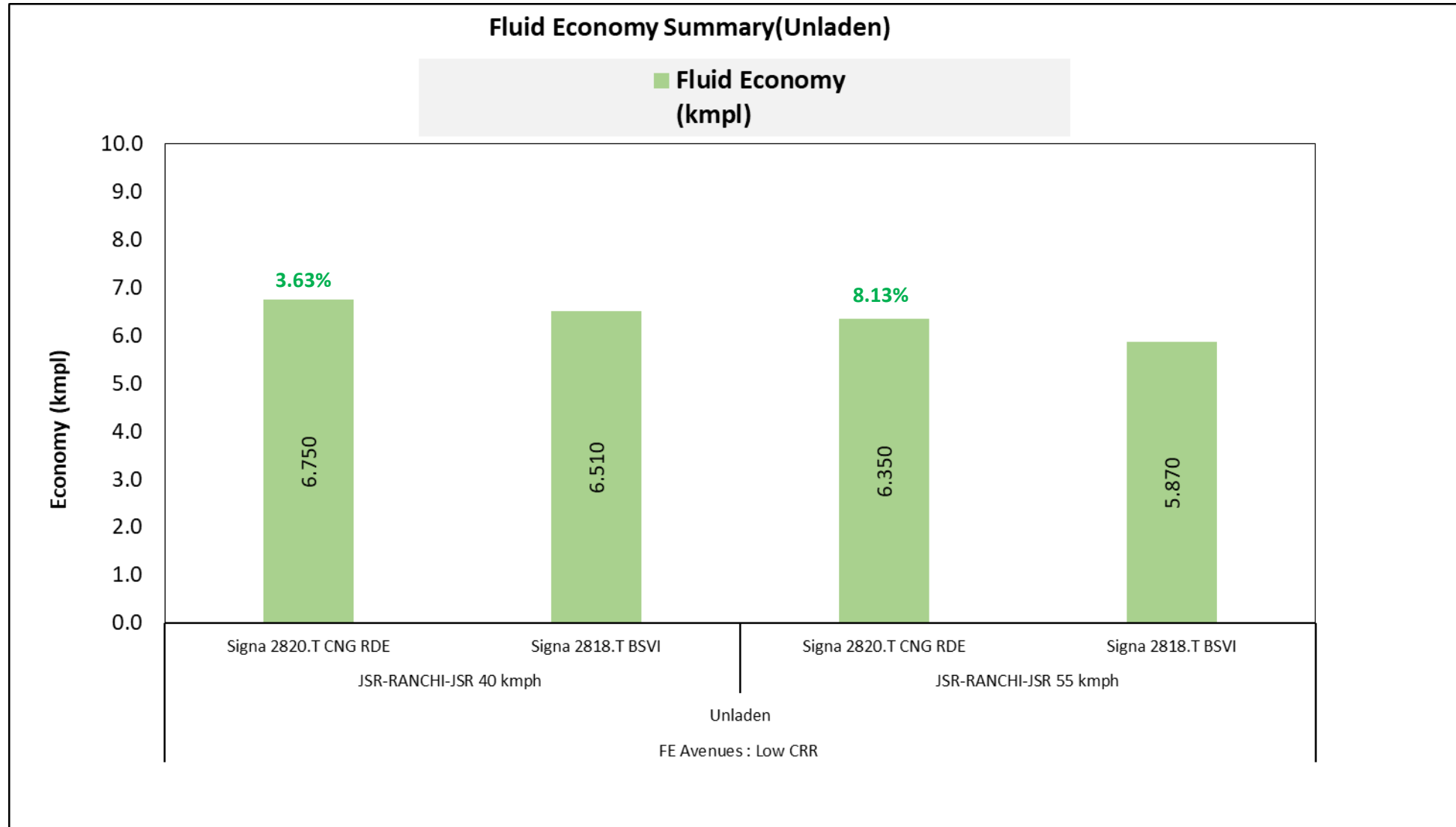
Annexure -3

Summary FE Status with all FE Avenues(Laden)



Summary FE Status with all FE Avenues(Un laden)

Annexure -3



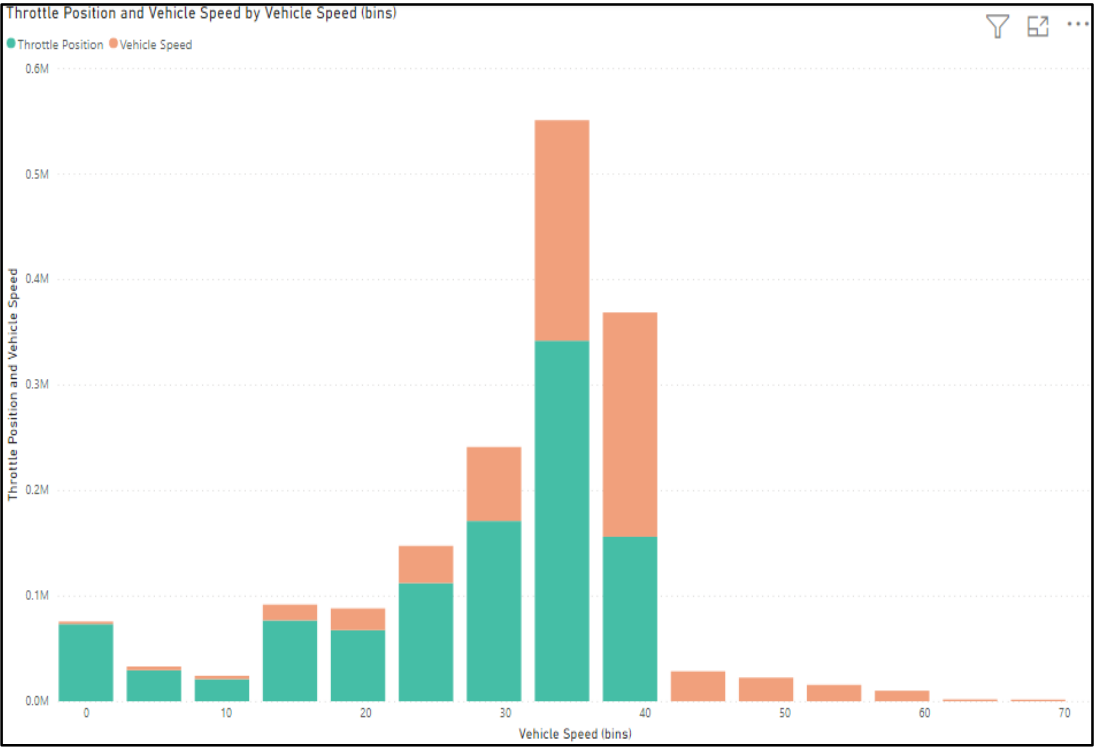
Duty Cycle Analysis
Jamshedpur-Ranchi Road FE Trial

Laden Trial

Signal 2820.T CNG BSVI RDE Test Vehicle vs Signal 2818.T

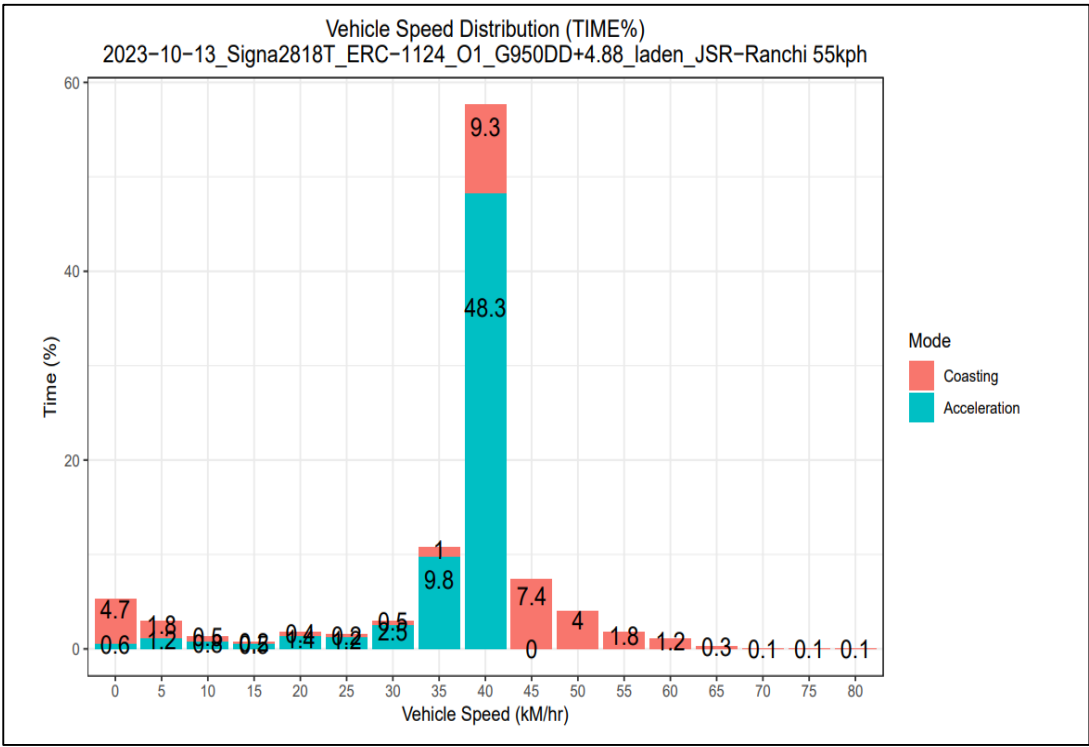
Test Vehicle

13.10.23 40 kmph



Base Vehicle

13.10.23 40 kmph

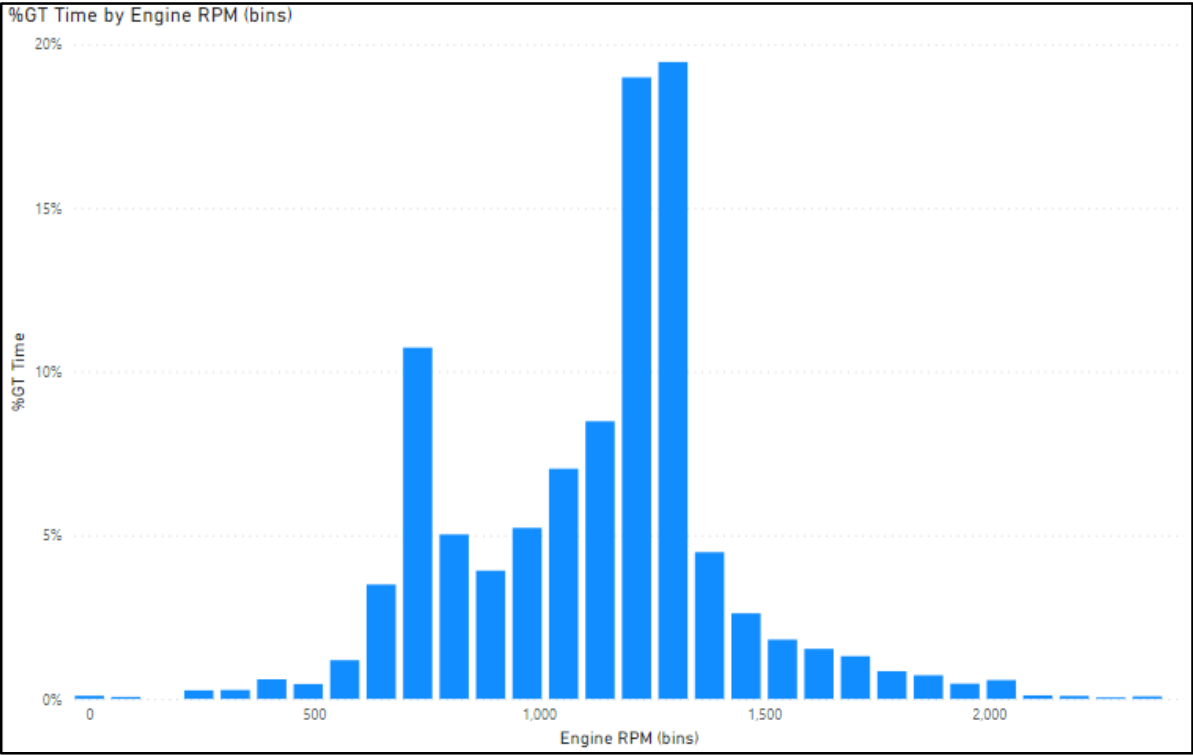


Laden Trial

Signa 2820.T CNG BSVI RDE Test Vehicle vs Signa 2818.T

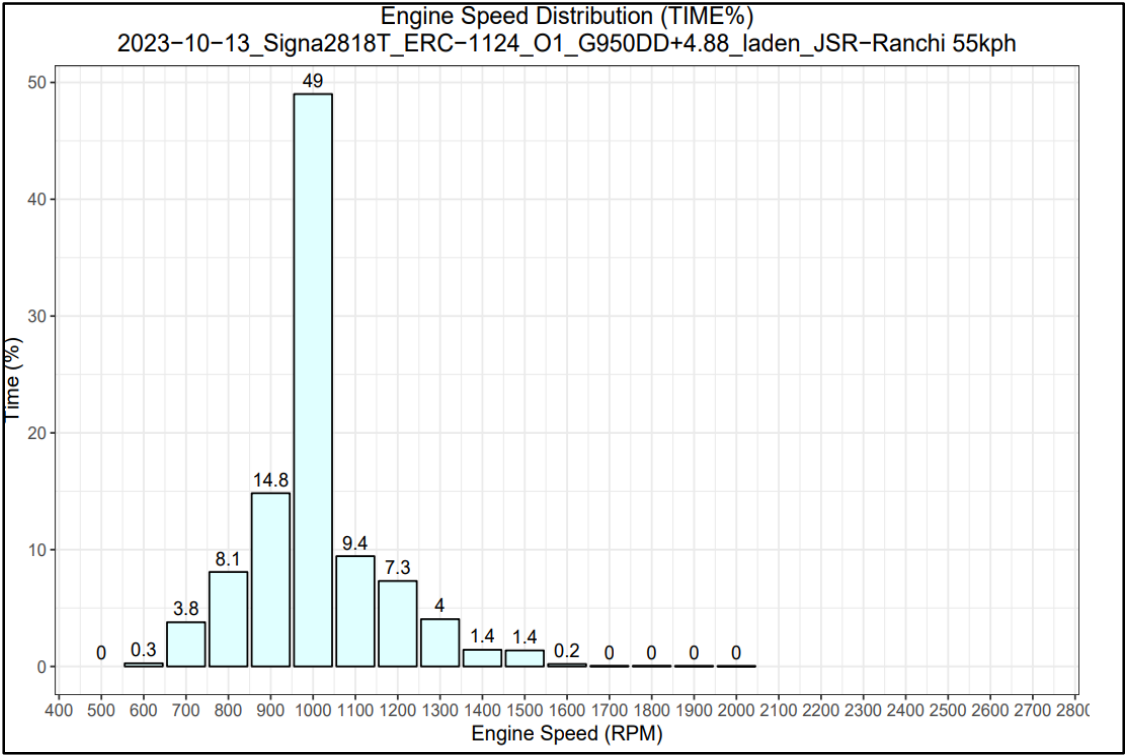
Test Vehicle

13.10.23 40 kmph



Base Vehicle

13.10.23 40 kmph

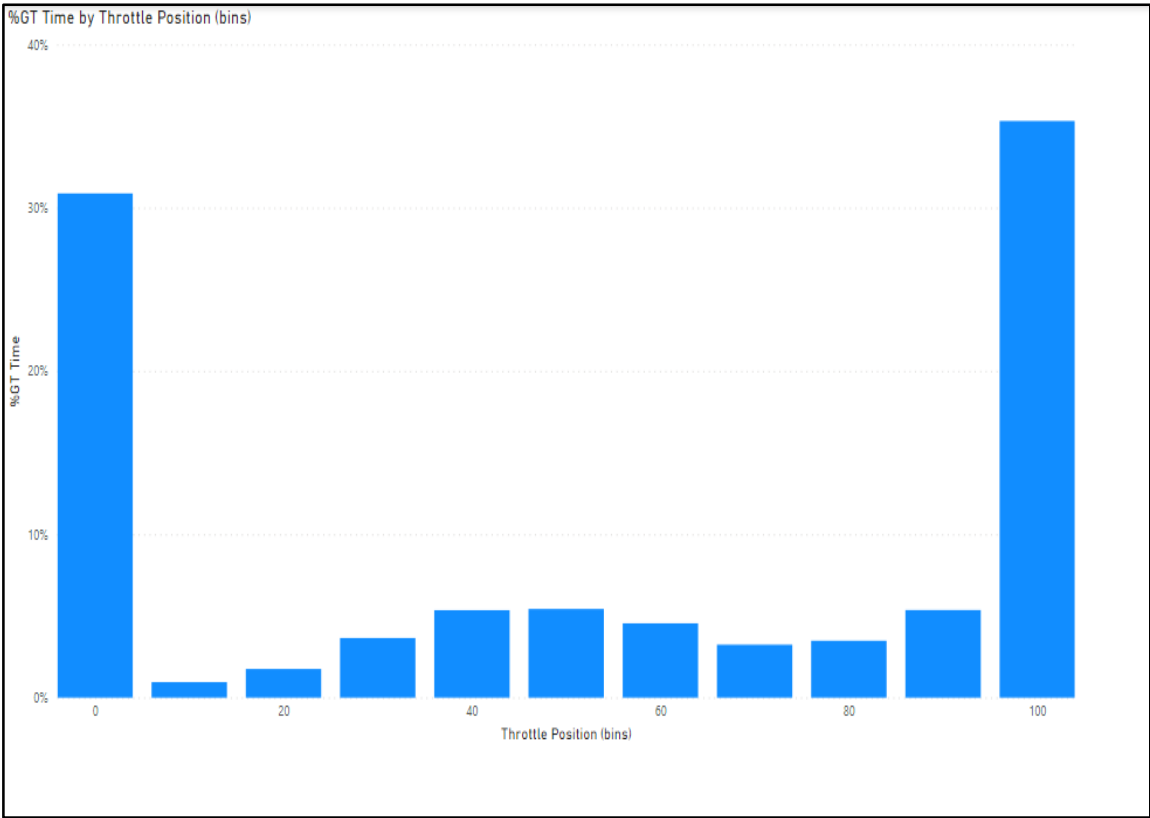


Laden Trial

Signal 2820.T CNG BSVI RDE Test Vehicle vs Signal 2818.T

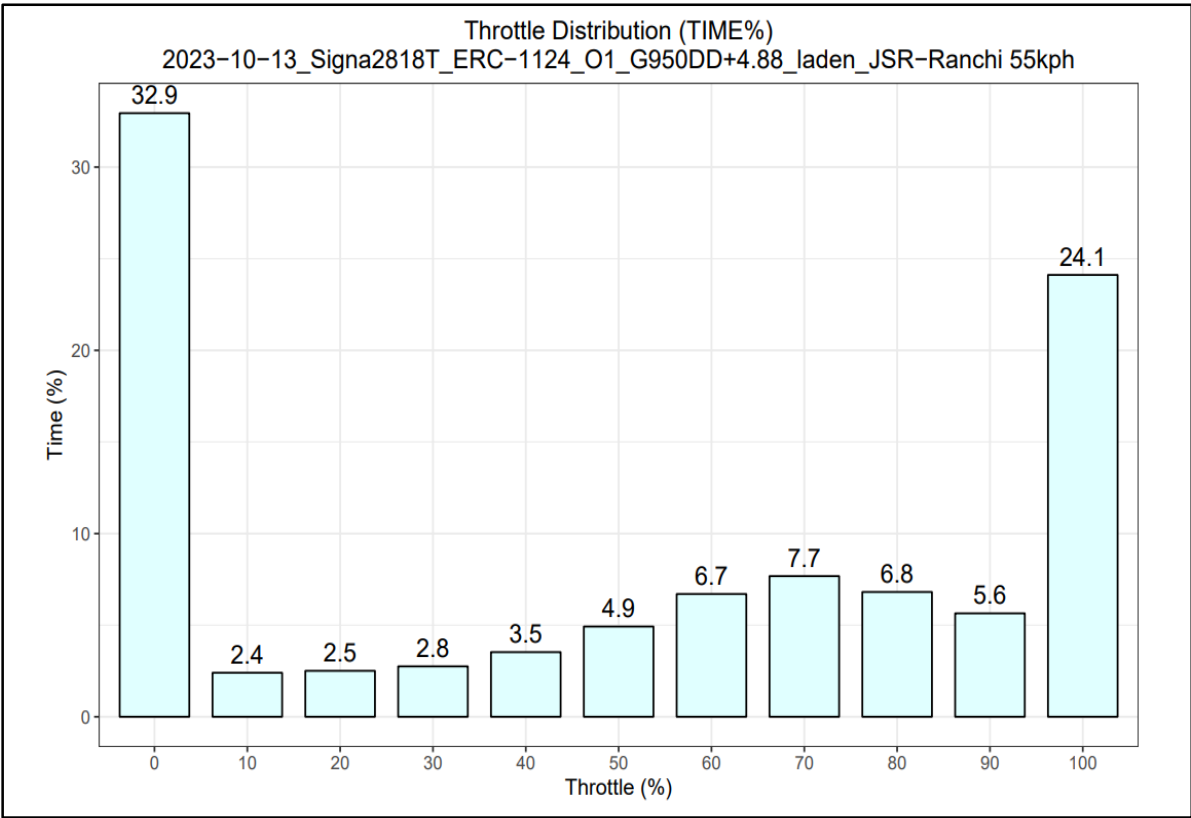
Test Vehicle

13.10.23 40 kmph



Base Vehicle

13.10.23 40 kmph

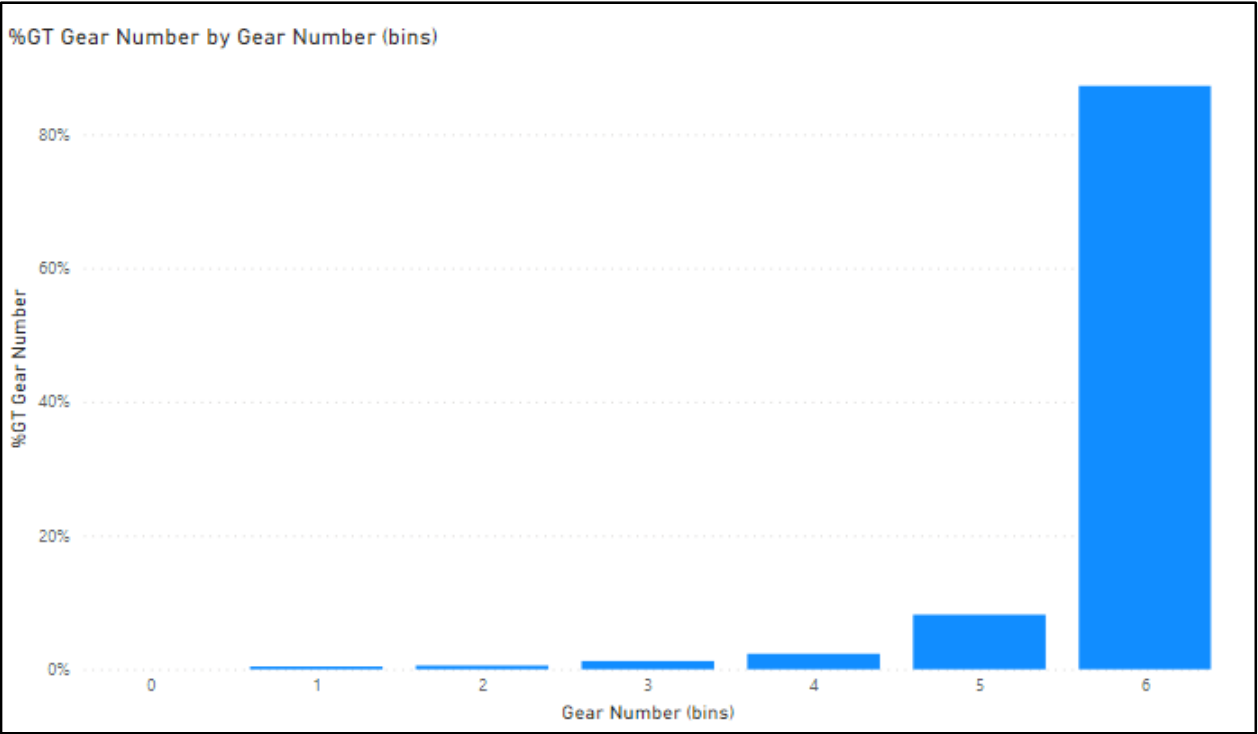


Laden Trial

Signal 2820.T CNG BSVI RDE Test Vehicle vs Signal 2818.T

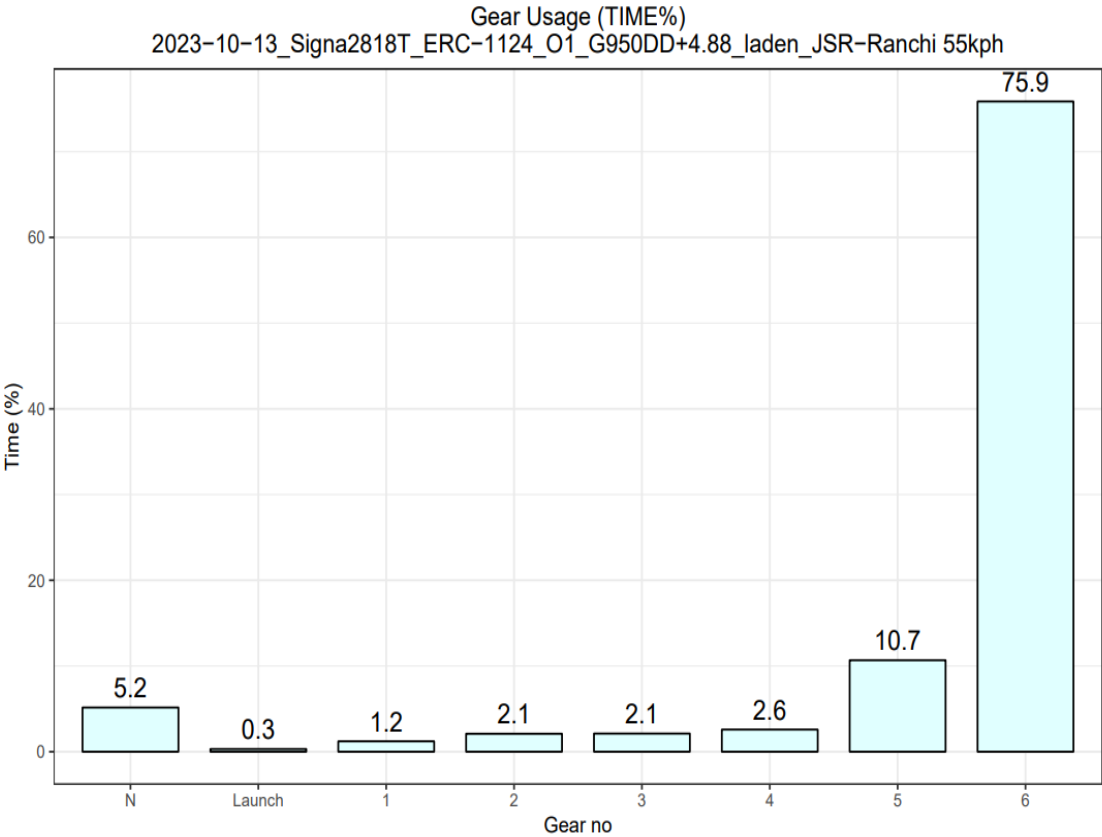
Test Vehicle

13.10.23 40 kmph



Base Vehicle

13.10.23 40 kmph

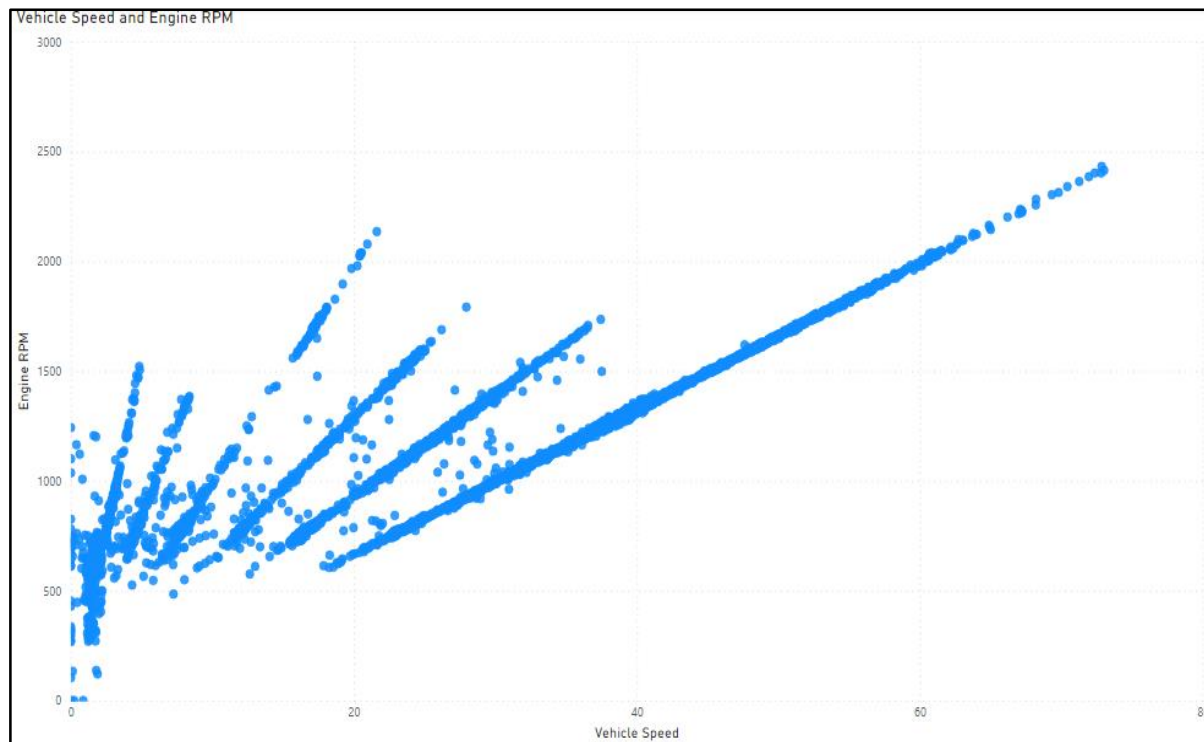


Laden Trial

Signa 2820.T CNG BSVI RDE Test Vehicle vs Signa 2818.T

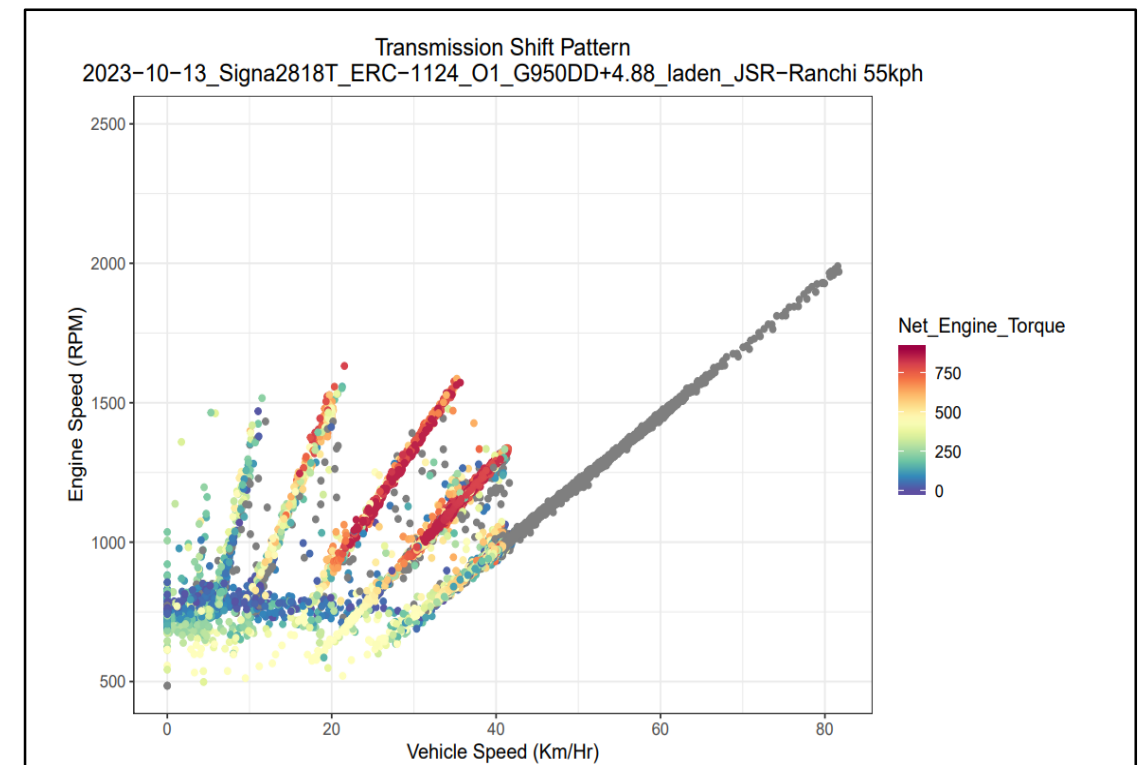
Test Vehicle

13.10.23 40 kmph



Base Vehicle

13.10.23 40 kmph

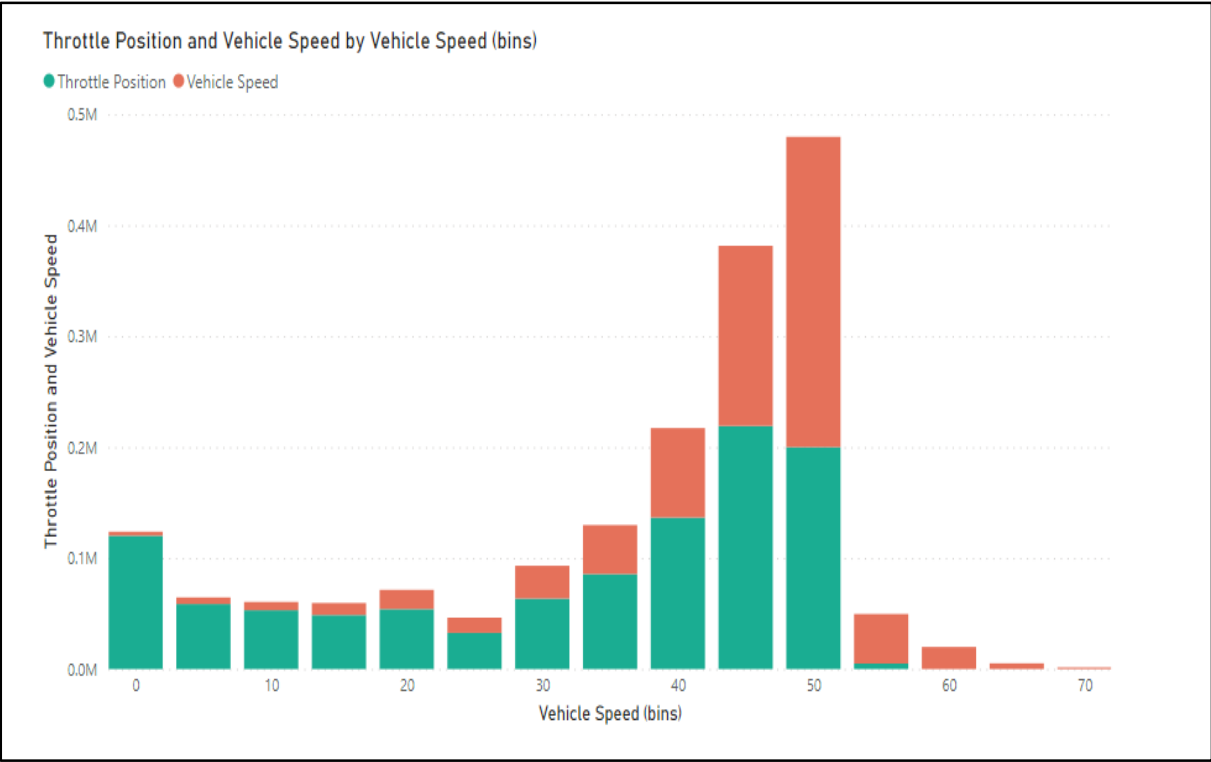


Laden Trial

Signal 2820.T CNG BSVI RDE Test Vehicle vs Signal 2818.T

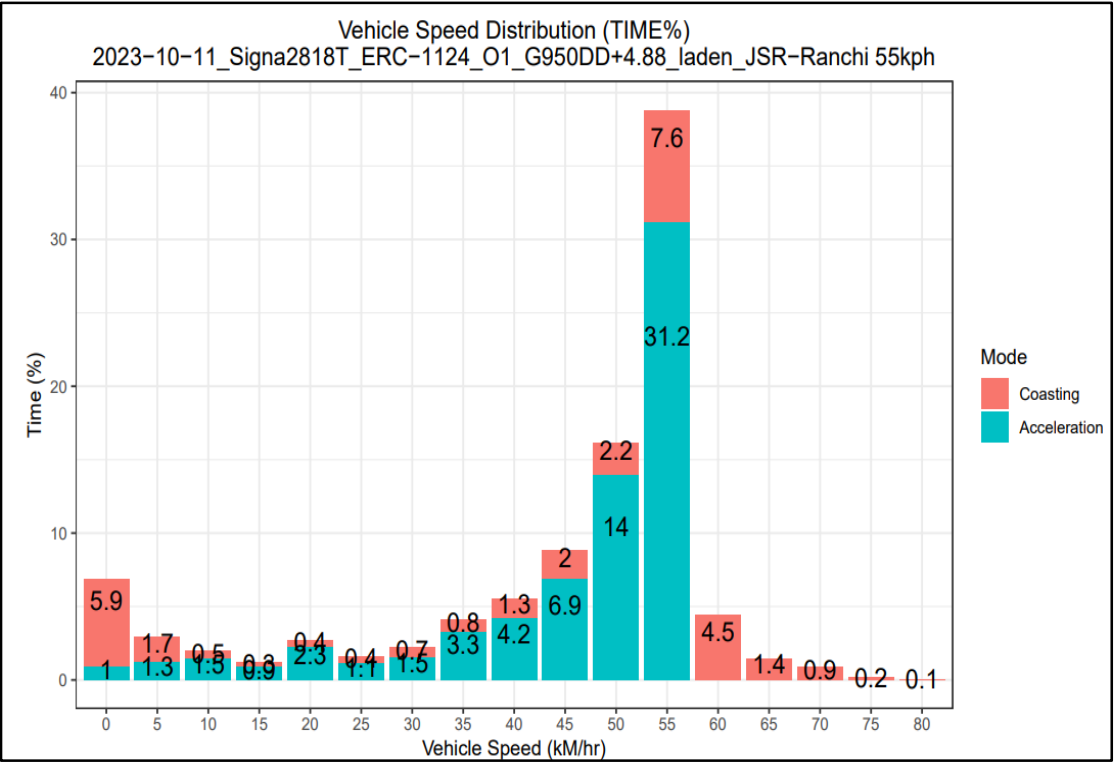
Test Vehicle

11.10.23 55 kmph



Base Vehicle

11.10.23 55 kmph

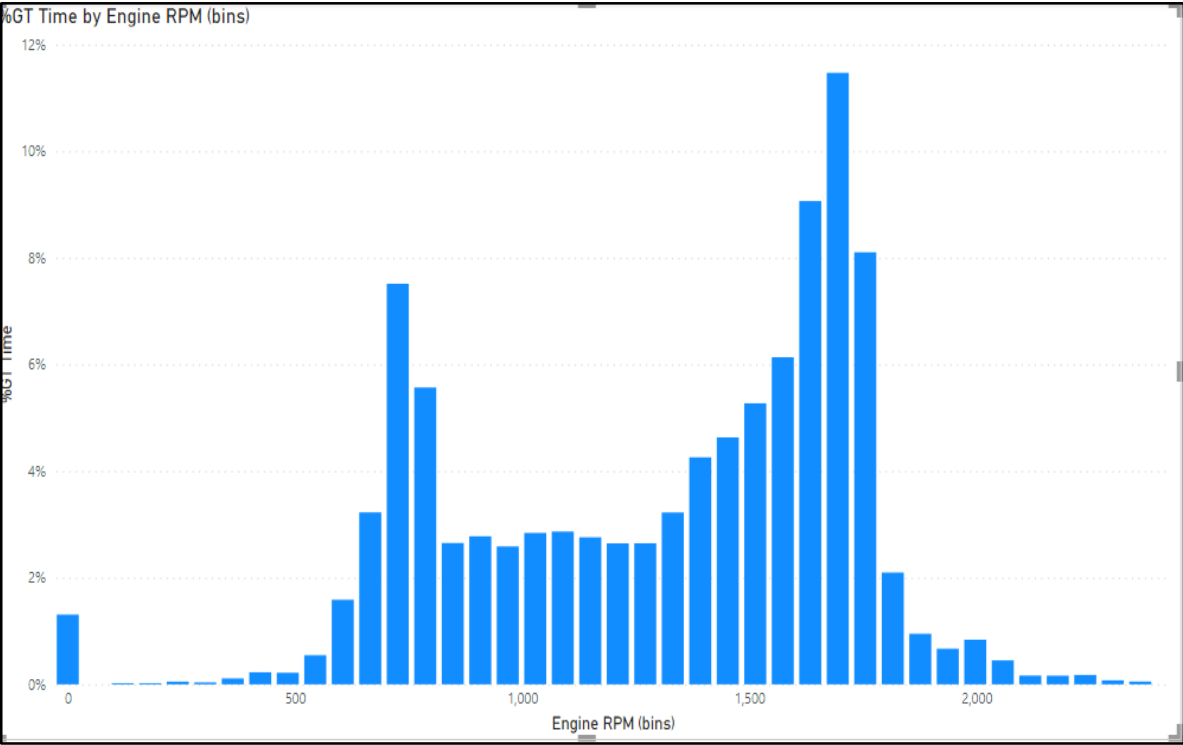


Laden Trial

Signal 2820.T CNG BSVI RDE Test Vehicle vs Signal 2818.T

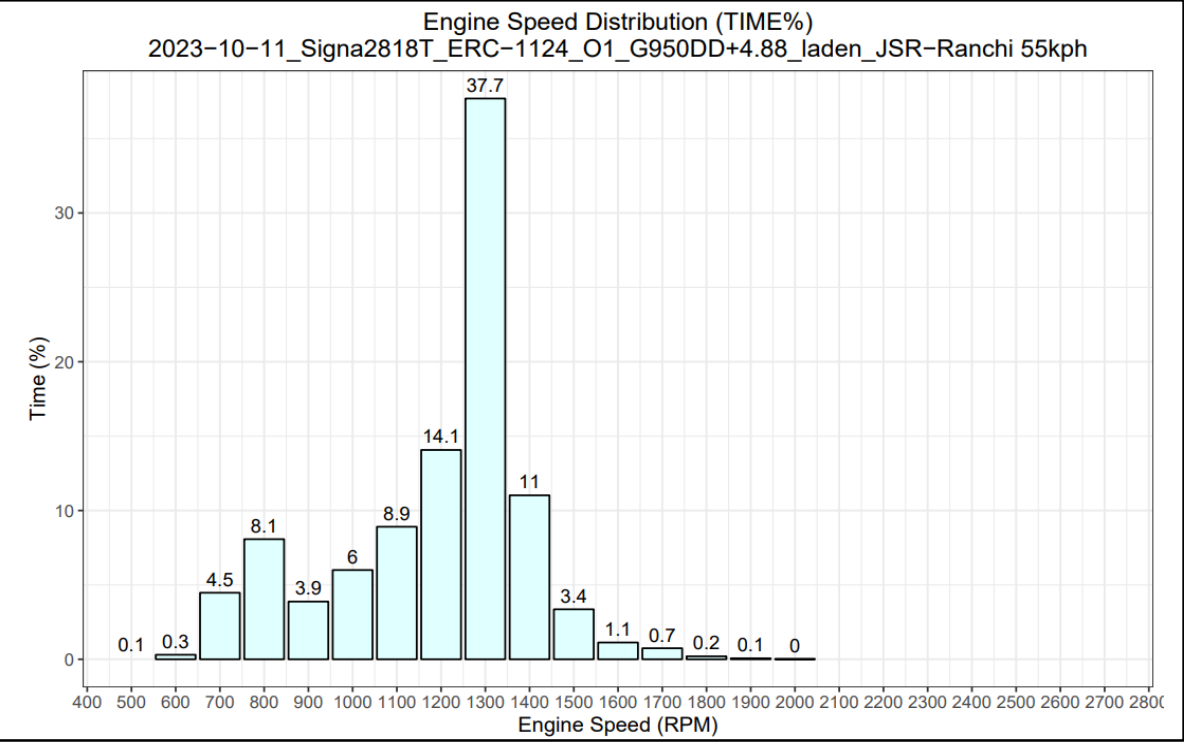
Test Vehicle

13.10.23 55 kmph



Base Vehicle

13.10.23 55 kmph

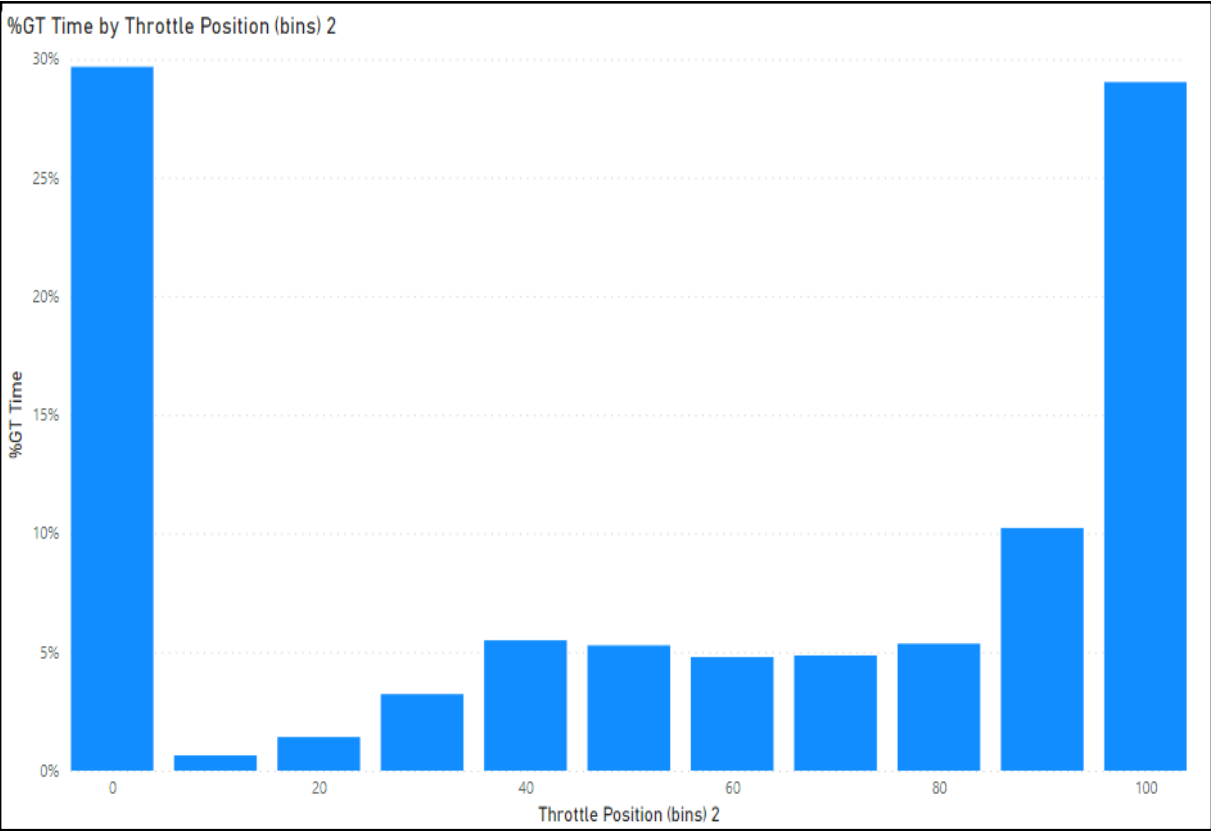


Laden Trial

Signal 2820.T CNG BSVI RDE Test Vehicle vs Signal 2818.T

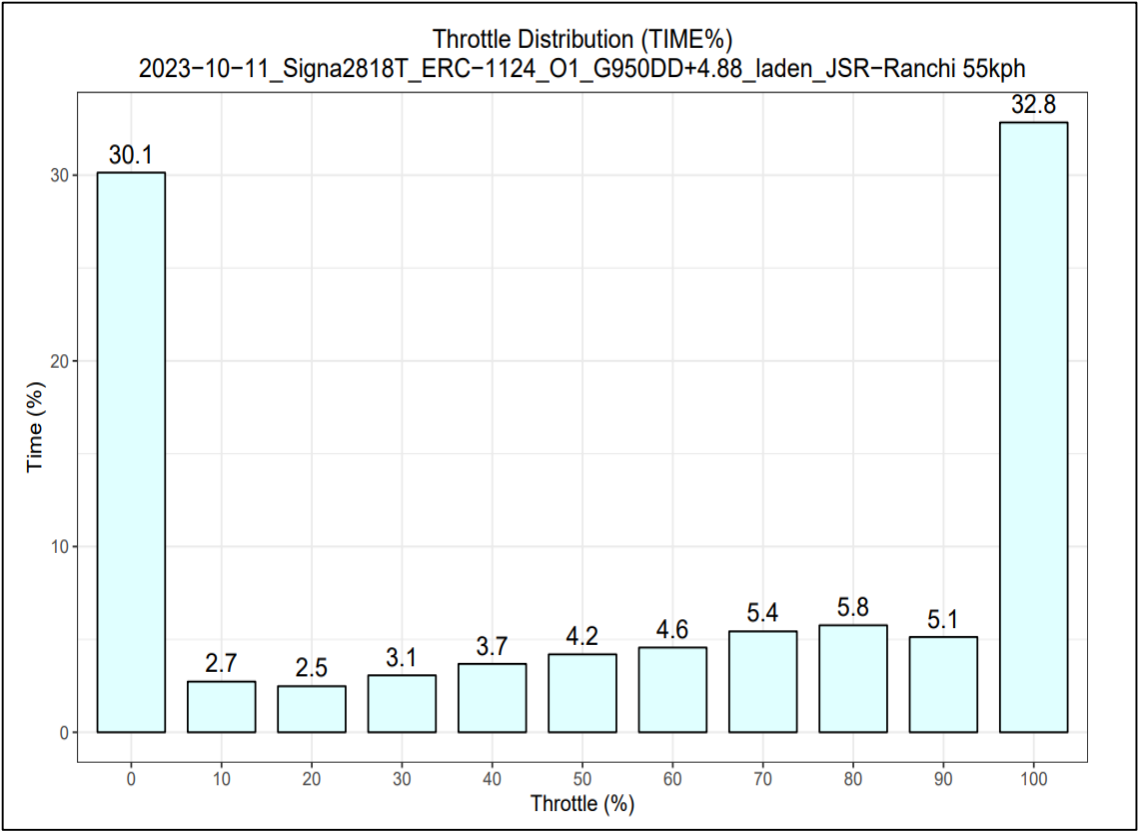
Test Vehicle

13.10.23 55 kmph



Base Vehicle

13.10.23 55 kmph

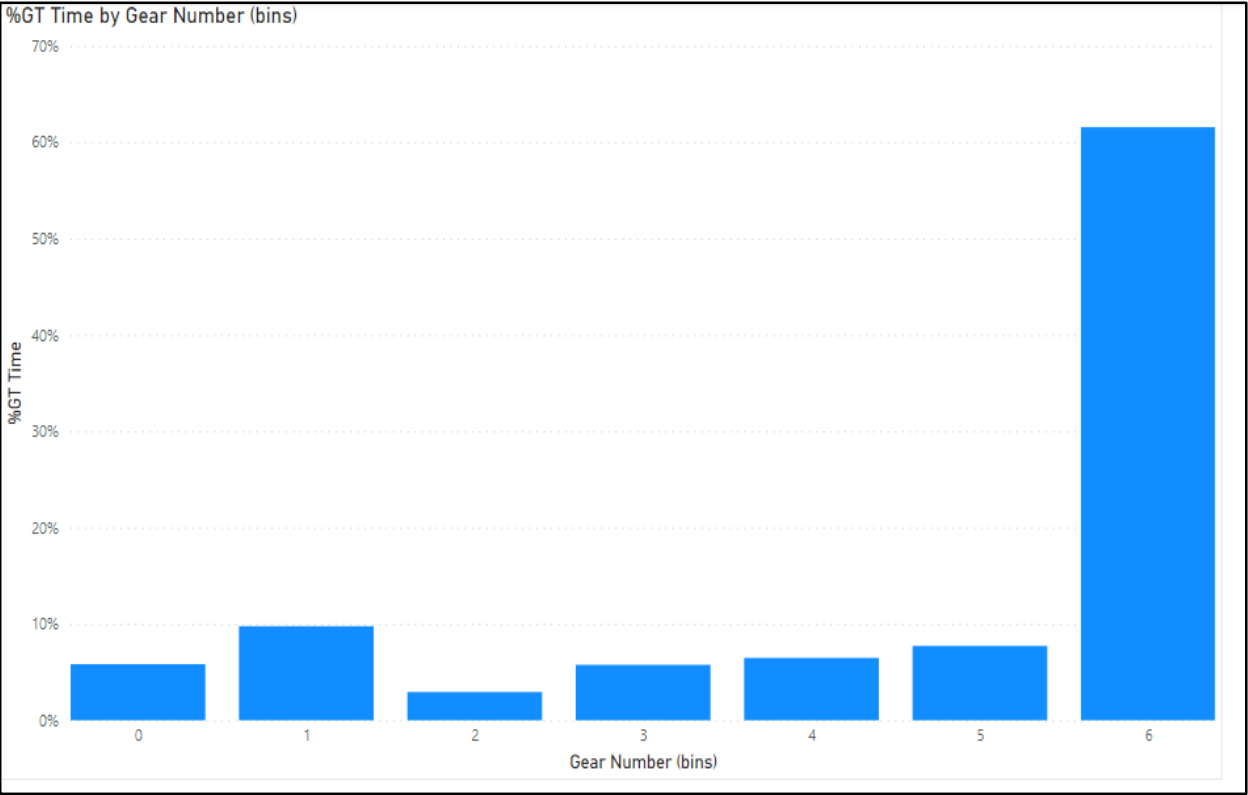


Laden Trial

Signal 2820.T CNG BSVI RDE Test Vehicle vs Signal 2818.T

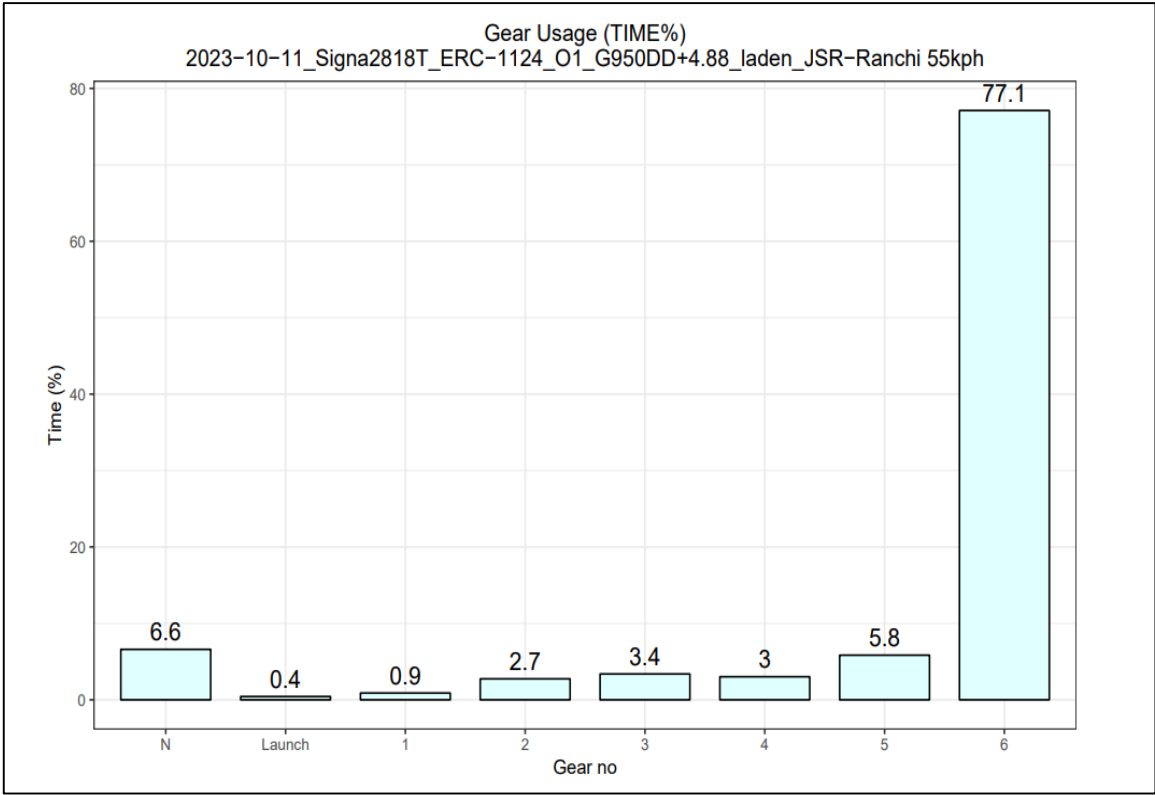
Test Vehicle

13.10.23 55 kmph



Base Vehicle

13.10.23 55 kmph

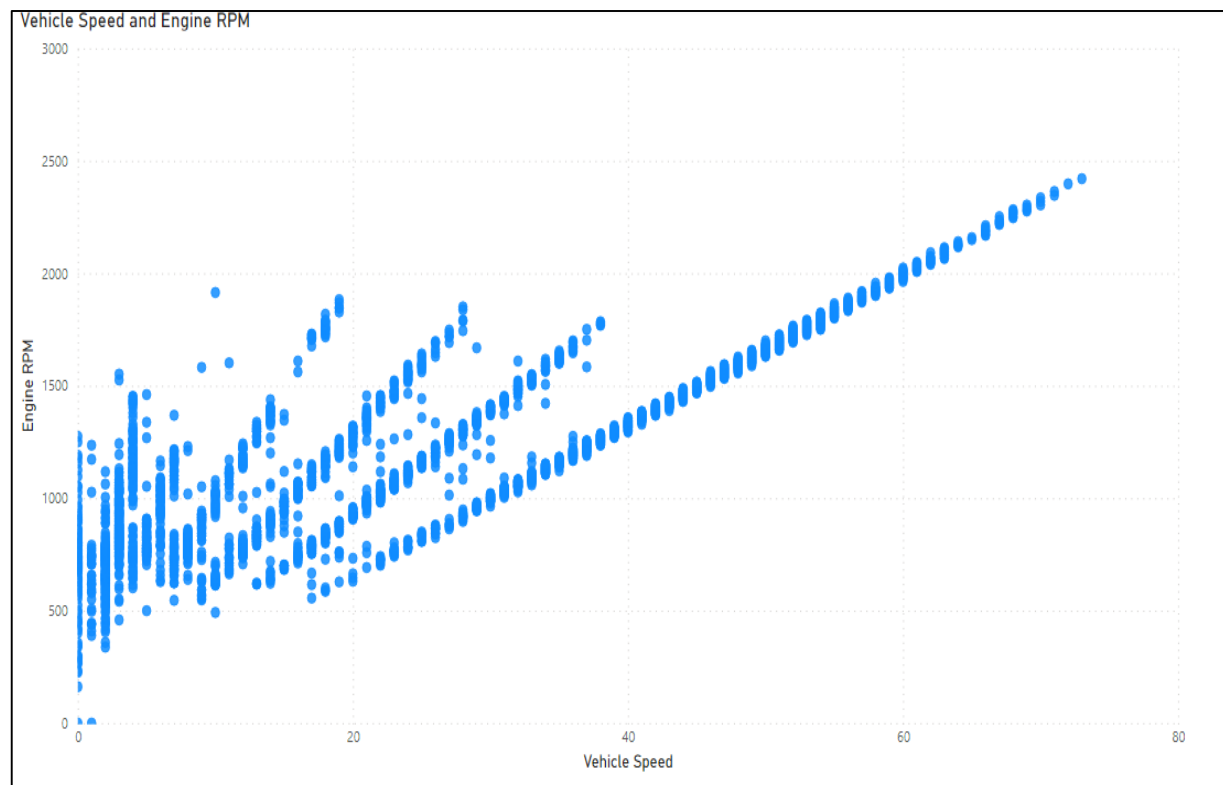


Laden Trial

Signa 2820.T CNG BSVI RDE Test Vehicle vs Signa 2818.T

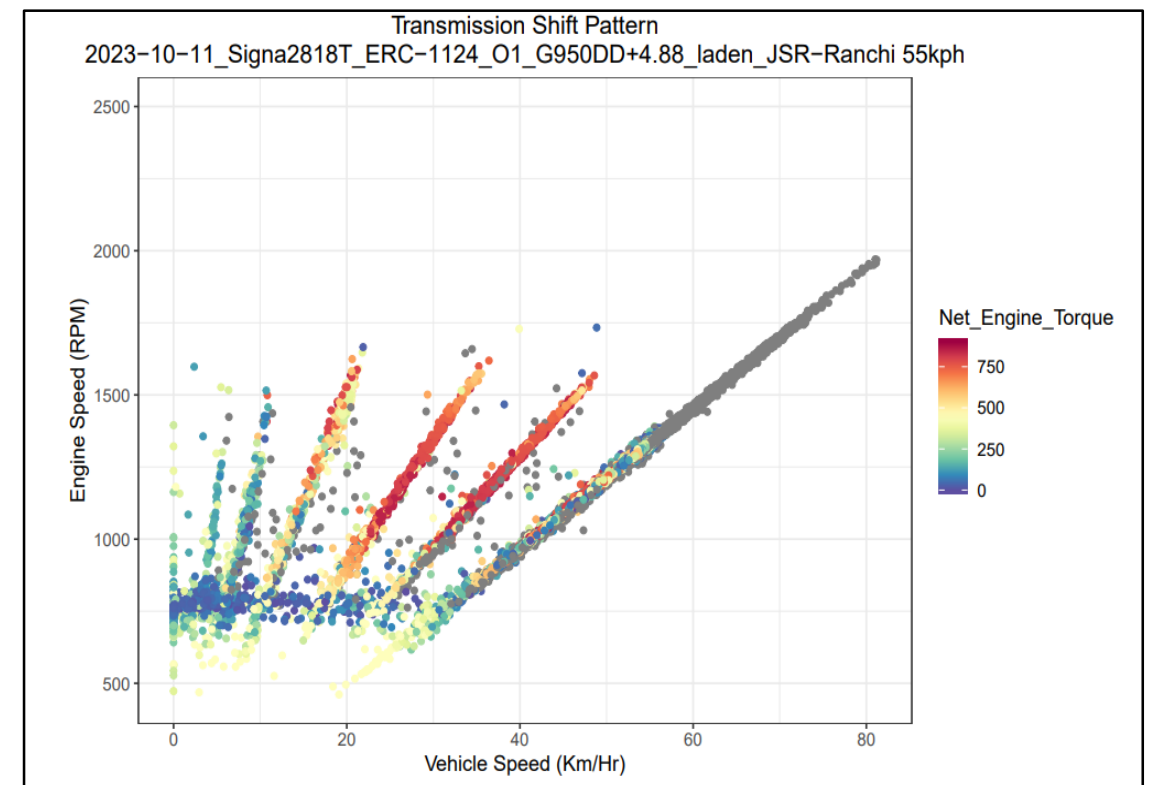
Test Vehicle

11.10.23 55 kmph



Base Vehicle

11.10.23 55 kmph



Fluid Economy(FLE) Status of 2820.T BSVI RDE Test vehicle w.r.t. 2818.T BSVI base vehicle :

1. Laden

- a) 40kmph :- Fluid Economy of 2821.T is **superior** than base vehicle by **13.17 %** .
- b) 55kmph :- Fluid Economy of 2821.T is **superior** than base vehicle by **32.28 %** .

2. Unladen

- a) 40kmph :- Fluid Economy of 2821.T is **superior** than base vehicle by **3.63 %**.
- b) 55kmph :- Fluid Economy of 2821.T is **superior** than base vehicle by **8.13 %**.

Overall Average Fluid Economy(FLE) Status of 2820.T BSVI RDE Test vehicle w.r.t. 2818.T BSVI base vehicle :

- a) **40kmph (Laden(80%) +Unladen(20%))**:- Fluid Economy of 2821.T is **superior** than base vehicle by **12 %** .
- b) **55kmph (Laden(80%) +Unladen(20%))**:- :- Fluid Economy of 2821.T is **superior** than base vehicle by **28 %** .

Thank You

TEST REQUISITION

Ref.		Date:	06/06/2023
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To,

ERC Testing

Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 2820.T BSVI CNG				
2	Part Number	51832155R,				
3	Drawing number	518300000014,	Mod status	D,2,	Mod Date	25/10/2023
4	Vehicle Model / Project	Signa 2820.T CNG RDe BSVI				
5	WBS Number	P.22.EA.497 -MHCV 5.7 SGI Truck BS6 RDE for 28T Truck				
6	Supplier	Tata Motrs				
7	Number of samples					
8	Inspection report	Enclosed	☐ Yes	☒ No	If No commitment date:	
9	Material report	Enclosed	☐ Yes	☒ No	If No commitment date:	
10	Suppliers test report	Enclosed	☒ Yes	☐ No	If No commitment date:	
11	Reason for testing	First Source Development	☐	Weight Reduction	☐	
		Alternate source development	☐	Cost Reduction	☐	
		Data generation Test	☐	Regulatory test	☐	
		Any other (pl. specify)	☐	New Development	☒	
12	What testing required?					
13	Remarks					
14	Peripheral Parts Availability	NO				

Authorized Signature:	
Name:	
Dept:	

Contact Person:	
Phone Number	

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

TATA MOTORS LIMITED ERC PUNE	TEST PLAN	DATE: 07/11/2023			
	VEHICLE TESTING DEPARTMENT				
Description: Signa 2820.T CNG RDe BSVI		Test Item No: 43448			
		New	☒	Moderate	☒
		Major	☒	Minor	☒

Part No: 51832155R,		Modification No: D,2,	Enclosures: Nil
Drawing No: 518300000014,			
Make: Tata Motrs		WBS Number: P.22.EA.497 -MHCV 5.7 SGI Truck BS6 RDE for 28T Truck	Copies to :
Vehicle Type: Signa 2820.T CNG RDe BSVI			
No. of Samples: 1			
Test Requested by:	M & HCV		Equipments: Nil
Test Requisition Reference:	Testing-Outdoor/15523		
Reason for Test/s:	New development		
Regulatory Requirement:	a) Component: Nil		
	b) Vehicle: Nil		

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	FE trial	FE trial	As per PALS	1	Debabrata Das	Beta	Nil

REMARKS: Test Plan for DVP		INTERNAL REF:	
TEST TO FAILURE: No			
MEOST: No			
RELIABILITY TESTING: No			
BENCHMARKING TESTING: No			
DFMEA/FMEA DONE: Yes			
PREPARED BY	CHECKED BY	APPROVED BY	